New experiments on 25 controlled models (DataDecide)

1 Overview

The paper claims that **how often numbers of each size appear in pretraining data** (α) is related to **how compressed the model's internal number line is** (β). The reviewers mainly asked whether other differences between the models explain this, whether 6 corpora are enough, whether the result is robust, and whether β is measured reliably. The three experiments below use a set of models built to remove most of these doubts.

Terms used throughout.

- **DataDecide.** 25 language models released by Ai2. All have 1B parameters, the same architecture (16 layers), the same tokenizer, and were trained for the same number of steps on 100B tokens. *Only the training data (the "recipe") differs.* Recipes come in **families**: variants of one base corpus (filtered subsets, versions with a part removed, or mixes). The families are Dolma, DCLM, Falcon, FineWeb, C4, and DCLM+Dolma mixes.
- α **(data exponent).** Count how often each whole number n appears in the training text, $c(n)$. Larger numbers are rarer. α is the slope of $\log c(n)$ against $\log n$. It is negative; a *less negative* α means large numbers are relatively more common.
- β **(number-line compression).** We give the model prompts that end in a number, take its internal state (hidden state) at one layer, and find the main direction along which those states vary (the first principal component, PC1). Numbers line up along PC1 by size. β says how the space between magnitude groups changes from one power of ten to the next: $\beta = 1$ gives each power of ten equal space (a logarithmic line); $\beta < 1$ squeezes large numbers even more. *Higher* β *= less compression.*
- **Prediction.** Flatter α (large numbers more common) $\Rightarrow$ higher β. So we expect a *positive* correlation between α and β.
- **Layer choice.** The number line is read from one layer per model, chosen by a fixed rule from the paper: the layer where PC1 explains the most variance *and* orders the numbers best (score $\sqrt{\text{explained variance} \times |\text{order}|}$).
- **Order.** Spearman correlation between n and its position on PC1 (1 = numbers perfectly in order).
- **Pearson** r **/ Spearman** ρ**.** r measures a straight-line relation; ρ only asks whether the *ranking* agrees and is less sensitive to single extreme models. p is the probability of seeing a correlation this strong if there were no real relation.

All results are correlations across models; none of them shows *causation.*

Table 1: The new experiments at a glance. Reviewer IDs refer to the AAAI reviews.

Exp.	Question	Main answer	Helps with
E01	Does the paper's result hold on 25 models that differ only in data?	Yes, moderately: $r = 0.56$ ($p = 0.003$), $\rho = 0.46$ ($p = 0.02$).	Confounding by architecture, tokenizer, size, training length; "only 6 corpora"; partial correlation for scale (H7EL, VRQv, a1nj).
E06	Is β reliable when fitted from only 3 gaps?	Yes: 10 groups, 3 seeds; β is stable and agrees with the 3-gap β ($r = 0.96$). Layer choice is the fragile part.	β from few points, fit quality, stability (a1nj pt. 3); sensitivity to the layer.
E11	What if α is measured on the same numbers β is measured on?	$r = 0.82$, $\rho = 0.77$; holds without Dolma, within families, and when any model is removed.	Sensitivity, leave-one-out, Dolma influence, by-family analysis, uncertainty (VRQv Q2, H7EL, a1nj Q1).

2 E01: The paper's method on 25 controlled models

Purpose. In the paper, models differ in architecture, tokenizer, size *and* data, so any of these could explain β. DataDecide fixes everything except the data.

What we did. Exactly the paper's protocol.

- Prompts `a=a,b=b,c=c,n=`: three examples of repeating a number, then the model must repeat n. Only prompts the model answers correctly are kept.
- n comes from 4 magnitude groups: 1–39, 80–119, 980–1019, 9980–10019 (40 prompts each). The 4 group positions on PC1 give **3 gaps**, from which β is fitted.
- The layer is chosen on one set of prompts (seeds 42–44) and β is measured on fresh prompts (seeds 45–47), so the layer is not tuned on the data it is evaluated on.
- α is the paper's α (slope over all numbers 1–10000), counted on the exact 100B tokens each model was trained on.

Results. Per-model values are in Table 7.

- Every model has a clear number line: order = 0.90–0.96.
- β ranges from 0.62 to 1.36. Dolma-trained models are least compressed.
- Mixing DCLM with more Dolma raises β step by step: 0.67, 0.84, 0.95, 1.11 for 0, 25, 50, 75% Dolma.

Table 2: α vs β: the paper versus the 25 DataDecide models.

Data	Points	Pearson r (p)	Spearman ρ (p)
Paper (corpus level, after inclusion audit)	6 corpora	0.81 (0.061)	0.60 (0.24)
DataDecide, E01	25 models	0.56 (0.003)	0.46 (0.022)

What it addresses.

- **Confounding by model design:** architecture, tokenizer, size and training length are identical, so they cannot explain the relation.
- **Too few points:** 25 models instead of 6 corpora; the correlation is weaker than the paper's but now statistically significant.
- **"Partial correlation controlling for scale":** unnecessary here, since scale is the same for all models.
- **Limitation:** the correlation is moderate and largely driven by the Dolma family (see E11).

3 E06: Is β reliable? (10 groups instead of 4)

Purpose. The paper fits β from only 3 gaps between 4 groups. Reviewers asked whether that is stable and how good the fit is.

What we did.

- 10 magnitude groups instead of 4, evenly spaced on a log scale (centres 10, 22, 46, 100, ..., 10,000, each $\pm 10\%$), giving 9 gaps.
- 100 correctly answered prompts per group, repeated with 3 random seeds (3,000 prompts per model).
- β is fitted with one smooth curve through all prompts, $\text{PC1} = a + s\,\beta^{\log_{10} n}$, instead of from group averages.
- We also report α vs β with *the same layer for every model*, to see how much the layer choice matters.

Table 3: Reliability of β (25 models).

Check	Result
Change in β across the 3 prompt seeds (median s.d.)	0.015
Spread of β between models	0.55–1.38
Fit quality of the smooth curve (R^2, median and range)	0.89 (0.81–0.97)
Agreement with E01's 3-gap β, same layer (Pearson r)	0.96
Models whose chosen layer changes vs E01	8 of 25

Results (reliability). β barely changes with new prompts, the curve fits well, and the paper's 3-gap β gives almost the same values. The fragile part is the *layer choice*: for 8 models the best and second-best layers score within 8% of each other, and switching between them can change β a lot (e.g. 0.81 at layer 9 vs 1.38 at layer 2 for one model). Because of these near-ties, α vs β is $r = 0.63$ ($p = 0.001$) at E01's layers but only 0.20 at E06's own choices.

Results (fixed layer). In the middle layers (6–11), where most models' chosen layers lie, α and β are clearly related at every layer ($r = 0.61$–0.75). In early layers (2–4) there is no relation.

What it addresses. "β is estimated from few points; how stable is it, how good is the fit?" (a1nj point 3): stable, good fit, and the 3-gap β agrees with the 10-group β. Also shows how the result depends on the layer, and that comparing models at the same depth gives a clear relation.

Table 4: α vs β when every model is read at the *same* layer (E06 measurement, original α, $n = 25$ per layer).

		Median	Pearson		Spearman	
Layer	Order	β	r	p	ρ	p
1†	0.07	0.20	+0.35	0.090	+0.32	0.117
2	0.90	0.93	−0.05	0.799	−0.03	0.878
3	0.96	0.65	−0.35	0.084	−0.37	0.069
4	0.96	0.69	−0.00	0.987	−0.09	0.677
5	0.96	0.65	+0.51	0.010	+0.30	0.149
6	0.95	0.64	**+0.68**	0.0002	+0.56	0.004
7	0.95	0.64	**+0.67**	0.0003	+0.53	0.007
8	0.94	0.66	**+0.75**	0.00002	**+0.71**	0.00008
9	0.94	0.71	**+0.64**	0.0006	+0.54	0.005
10	0.94	0.72	**+0.61**	0.001	+0.45	0.023
11	0.94	0.70	**+0.66**	0.0004	**+0.59**	0.002
12	0.89	0.63	+0.48	0.016	+0.49	0.014
13†	0.44	0.53	−0.37	0.065	−0.14	0.504
14†	0.30	0.28	+0.26	0.214	+0.27	0.184
15†	0.30	0.21	−0.17	0.416	−0.20	0.349
16†	0.57	0.40	−0.35	0.085	+0.12	0.553

Order: median over models of how well PC1 orders the numbers. †Order below 0.85: no clear number line at that layer, so β there is not meaningful. **Bold**: significant after correcting for testing 16 layers ($p < 0.05/16 = 0.003$).

4 E11: α measured on the numbers that β uses

Purpose. β is measured only on 4 groups of numbers (1–39, 80–119, 980–1019, 9980–10019). The original α is a slope over *all* numbers 1–10000, so 90% of its points are 4-digit numbers (1000–9999) that β never sees. The two measures therefore describe different numbers. E11 measures α on the same numbers as β.

What we did.

- α_{old}: slope of $\log c(n)$ against $\log n$ over every number 1–10000 (as in the paper).
- α_{new}: for each of the 4 groups, take the median count of its numbers and the group's centre; α_{new} is the slope of log(median count) against log(centre), i.e. 4 points, one per group. The median keeps one unusually common number (e.g. 10000) from dominating a group. Counts stop at 10000, so the top group uses 9980–10000.
- Same 25 models, same β and layers as E01. No model is dropped.

Table 5: Example: why the two α can disagree. Median count of the numbers in each group.

Model	1–39	80–119	980–1019	9980–10000	α_{old}	α_{new}	β
`fineweb-edu`	13.4M	1.29M	51k	1.9k	−1.70	−1.37	**0.89**
`c4`	11.4M	0.93M	34k	1.5k	−1.56	−1.39	**0.62**

α_{old} says fineweb-edu's numbers become rare faster, so it should be *more* compressed; it is less compressed. On the probed numbers, fineweb-edu's counts actually drop more slowly, and α_{new} ranks the two correctly. The difference in α_{old} comes from 4-digit numbers between the probed groups.

Table 6: α vs β with both definitions of α. "Within families": each family's average removed first. $^*p < 0.05$. †`dolma1_7-no-reddit` (its β does not replicate on new prompts) and `dolma1_7`, `fineweb-edu`, `falcon-and-cc-qc-10p` (two layers score almost equally but give different β).

		α_{old} (1–10000)		α_{new} (probed numbers)	
Models used	n	r	ρ	r	ρ
All 25	25	0.56*	0.46*	**0.82***	**0.77***
Without 4 models with measurement doubts†	21	0.50*	0.38	0.79*	0.70*
Without the 6 Dolma models	19	0.22	0.11	0.68*	0.66*
Without all 9 Dolma-containing models	16	−0.31	−0.33	0.43	0.53*
Without the 8 recipes with code or math	17	−0.38	−0.41	0.44	0.52*
Without the 7 DCLM models	18	0.59*	0.59*	0.86*	0.82*
Without the 6 Falcon models	19	0.63*	0.62*	0.83*	0.81*
Only the 8 base corpora (one per original corpus)	8	0.55	0.50	0.92*	0.90*
Within families	25	0.28	0.24	0.75*	0.75*
Remove one model at a time (range)	24	0.50–0.61	0.39–0.52	0.79–0.84	0.75–0.80

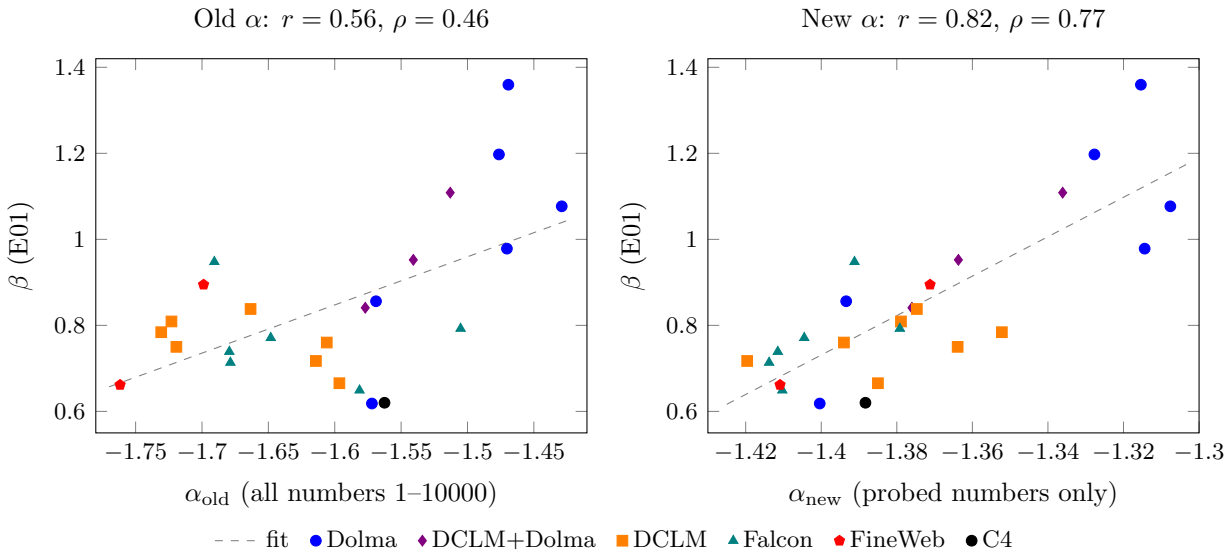


Figure 1: β against the two definitions of α; one point per model, colour = family. With the old α the trend comes mostly from Dolma (blue) versus the rest; with the new α the web-only families also follow the trend.

Results.

- With α_{new} the relation is much stronger: $r = 0.82$ ($p < 10^{-6}$), $\rho = 0.77$, with no model removed.
- It does not depend on any single model (leave-one-out: $r = 0.79$–0.84) or family.
- It survives removing Dolma ($r = 0.68$), holds within families ($r = 0.75$, vs 0.28 for α_{old}), and is strongest on the 8 independent base corpora ($r = 0.92$).
- The recipes are related (variants of the same corpus), so we also resampled whole families (family-level bootstrap): 95% interval for r is $[0.39, 0.90]$, which excludes 0. Using one average point per family: $r = 0.84$ ($p = 0.035$, 6 points).
- Controlling for code+math share weakens it: partial $r = 0.30$ ($p \approx 0.15$) for α_{new}, -0.23 for α_{old}. Code/math content explains part of the relation.

Caveats.

- **Chosen after seeing the data.** This α was adopted after the original α was known to correlate weakly. A second natural variant (pooling every probed number) agrees ($r = 0.79$, $\rho = 0.80$). It should be reported as exploratory and next to α_{old}, until tested on new models.
- Only 4 points per model, so α_{new} is a coarse slope.
- Without any code/math recipes the relation is positive but moderate ($\rho = 0.52$, $r = 0.44$, $p = 0.08$): the content confound is reduced, not removed.
- With E06's β at E06's own layer choices the relation is weaker ($r = 0.60$), for the near-tie reason in E06; at E01's layers it is $r = 0.77$. At a fixed layer 8 the new α is no better than the old one ($r = 0.75$ for both).

What it addresses. Sensitivity to individual corpora, leave-one-out, the influence of Dolma, analysis by family, and uncertainty (VRQv Q2, H7EL, a1nj Q1). It also explains *why* the original α is weak: it measures mostly numbers β does not probe.

Appendix: all 25 models

Table 7: Per-model values, sorted by β. Layer and Order from E01; β (E01) $\pm$ s.d. over 3 prompt seeds; β (E06) = 10-group smooth fit at the same layer. *Recipe contains code or math documents.

Model	Family	Layer	Order	β (E01)	β (E06)	α_{old}	α_{new}
dolma1_7-no-reddit*	Dolma	8	0.91	1.36 ± 0.06	1.18	-1.47	-1.315
dolma1_7-no-flan*	Dolma	9	0.93	1.20 ± 0.06	1.19	-1.48	-1.328
dclm-baseline-25p-dolma1.7-75p*	DCLM+Dolma	8	0.93	1.11 ± 0.06	1.10	-1.51	-1.336
dolma1_6plus*	Dolma	9	0.92	1.08 ± 0.06	1.01	-1.43	-1.308
dolma1_7*	Dolma	6	0.93	0.98 ± 0.08	0.92	-1.47	-1.314
dclm-baseline-50p-dolma1.7-50p*	DCLM+Dolma	8	0.95	0.95 ± 0.05	0.83	-1.54	-1.364
falcon-and-cc-qc-tulu-10p	Falcon	9	0.93	0.95 ± 0.04	0.81	-1.69	-1.391
fineweb-edu	FineWeb	9	0.96	0.89 ± 0.04	0.87	-1.70	-1.371
dolma1_7-no-code*	Dolma	8	0.94	0.86 ± 0.06	0.77	-1.57	-1.393
dclm-baseline-75p-dolma1.7-25p*	DCLM+Dolma	10	0.94	0.84 ± 0.07	0.76	-1.58	-1.376
dclm-baseline-qc-7p-fw2	DCLM	8	0.95	0.84 ± 0.04	0.67	-1.66	-1.375
dclm-baseline-qc-fw-10p	DCLM	3	0.94	0.81 ± 0.03	0.76	-1.72	-1.379
falcon	Falcon	9	0.92	0.79 ± 0.05	0.68	-1.51	-1.379
dclm-baseline-qc-fw-3p	DCLM	9	0.93	0.78 ± 0.00	0.57	-1.73	-1.352
falcon-and-cc-qc-orig-10p	Falcon	8	0.94	0.77 ± 0.06	0.66	-1.65	-1.405
dclm-baseline-qc-20p	DCLM	9	0.95	0.76 ± 0.06	0.72	-1.61	-1.394
dclm-baseline-qc-7p-fw3	DCLM	9	0.92	0.75 ± 0.08	0.60	-1.72	-1.364
falcon-and-cc-qc-20p	Falcon	8	0.93	0.74 ± 0.06	0.68	-1.68	-1.411
dclm-baseline-qc-10p	DCLM	8	0.93	0.72 ± 0.06	0.66	-1.61	-1.420
falcon-and-cc-qc-10p	Falcon	8	0.92	0.71 ± 0.07	0.60	-1.68	-1.414
dclm-baseline	DCLM	8	0.90	0.67 ± 0.07	0.58	-1.60	-1.385
fineweb-pro	FineWeb	7	0.93	0.66 ± 0.01	0.58	-1.76	-1.411
falcon-and-cc	Falcon	9	0.93	0.65 ± 0.04	0.63	-1.58	-1.410
c4	C4	7	0.93	0.62 ± 0.05	0.57	-1.56	-1.388
dolma1_7-no-math-code	Dolma	4	0.94	0.62 ± 0.04	0.55	-1.57	-1.400